

# MS Algebra Recitation 2

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## 1 Good Group Actions to Know

The following are group actions. You should verify this.

- Let  $G$  be a group and  $H \leq G$ . Then  $G \curvearrowright G/H$  via  $g_1 \cdot g_2H = (g_1g_2)H$ .
- Let  $G$  be a group,  $g \in G$ , and define  $\varphi_g(h) = ghg^{-1}$ . Then  $G \curvearrowright G$  via  $g \cdot h = \varphi_g(h)$ . Moreover,  $\varphi_g$  is an automorphism of  $G$ , and  $\text{im } \varphi \trianglelefteq \text{Aut}(G)$ .
- Let  $G$  be a group. Then  $G \curvearrowright \mathcal{P}(G)$  via  $g \cdot X = gX$ .
- Let  $G$  be a group. Then  $G \curvearrowright \mathcal{P}(G)$  via  $g \cdot x = gXg^{-1}$ .
- $S_n \curvearrowright [n]$  via  $\sigma \cdot x = \sigma(x)$ .
- $\mathbb{Z}/n\mathbb{Z} \curvearrowright [n]$  via  $k \cdot x = x + k \pmod n$ . This is the same as the action of  $\mathbb{Z}/n\mathbb{Z}$  on itself by left multiplication.
- For  $k$  a field,

$$\text{GL}_n(k) = \{A \in k^{n \times n} : \det A \neq 0\}$$

acts on  $k^n$  via  $A \cdot v = Av$ .

## 2 Finding Normal Subgroups

We first prove an equivalence of normal subgroups.

**Lemma.** *Let  $G$  be a group,  $g \in G$ , and  $X, Y \subseteq G$ . Then  $X \subseteq Y$  if and only if  $gX \subseteq gY$  if and only if  $Xg \subseteq Yg$ . In particular,  $X = Y$  if and only if  $gX = gY$  if and only if  $Xg = Yg$ .*

*Proof.* We show that  $X \subseteq Y$  if and only if  $gX \subseteq gY$ . If  $X \subseteq Y$ , then

$$gX = \{gx : x \in X\} \subseteq \{gy : y \in Y\} = gY.$$

Conversely, if  $gX \subseteq gY$ , then this argument shows that  $g^{-1}(gX) \subseteq g^{-1}(gY)$ . But  $g^{-1}(gX) = \{g^{-1}x' : x' \in gX\} = \{g^{-1}gx : x \in X\} = X$  and similarly for  $Y$ . ■

**Proposition.** *Let  $G$  be a group and  $H \leq G$ . The following are equivalent:*

- (1) *For any  $g \in G$ ,  $gH = Hg$ .*
- (2) *For any  $g \in G$ ,  $gHg^{-1} = H$ .*
- (3) *For any  $g \in G$ ,  $gHg^{-1} \subseteq H$ .*

*When any of these hold, we say  $H$  is normal in  $G$  and write  $H \trianglelefteq G$ .*

*Proof.*

(1) $\Leftrightarrow$ (2) By the lemma,  $gH = Hg$  if and only if  $gHg^{-1} = Hgg^{-1} = H$ .

(2) $\Rightarrow$ (3) Clear.

(3) $\Rightarrow$ (2) We have that  $gHg^{-1} \subseteq H$ . We also have that  $g^{-1}Hg = g^{-1}H(g^{-1})^{-1} \subseteq H$ . Multiplying on the left by  $g$  and on the right by  $g^{-1}$ ,  $H \subseteq gHg^{-1}$ , so we have equality.

■

Using these equivalent formulations, let's find some normal subgroups.

**Proposition.** *Let  $G$  be a group and  $H \leq G$  with  $[G : H] = 2$ . Then  $H \trianglelefteq G$ .*

*Proof.* Let  $g \in G$ . If  $g \in H$ , then  $gH = H = Hg$  and we are done. Otherwise,  $gH \neq H$ . Since the cosets of  $H$  partition  $G$ , we have that  $G = H \sqcup gH$  and  $G = H \sqcup Hg$ . But then  $gH = Hg$  anyways. ■

**Proposition.** *Let  $G$  be a finite group and let  $p$  be the least prime dividing  $|G|$ . If  $H \leq G$  with  $[G : H] = p$  then  $H \trianglelefteq G$ .*

*Proof.* Let  $X$  be the set  $G/H$  of left cosets of  $H$ , and let  $H \curvearrowright X$  by left multiplication (that is,  $h \cdot gH = (hg)H$ ).

For any coset  $gH$  of  $H$ , consider the orbit  $\mathcal{O}_{gH}$ . By orbit-stabilizer,  $|\mathcal{O}_{gH}| = [H : G_{gH}]$  divides  $|H|$ , and so divides  $|G|$ . However, we also know that  $|\mathcal{O}_{gH}| \leq |X| = p$ . Since  $p$  is the least prime dividing  $|G|$ , this means  $|\mathcal{O}_{gH}| = 1$  or  $p$ . Moreover, since this is true of all orbits, either we have one orbit of size  $p$ , or  $p$  disjoint orbits of size 1.

Now, for  $h \in H$ ,  $h \cdot H = H$ , so that  $|\mathcal{O}_H|$  has size 1. Thus, we must have  $p$  disjoint orbits of size 1. This means that, for any  $g \in G$  and  $h \in H$ ,  $hgH = gH$ , so that  $g^{-1}hg \in H$ . Thus,  $H \trianglelefteq G$ . ■

### 3 Classifying Homomorphisms

The goal of this section is to classify maps without appealing to “there exists an element such that...”. That is, we are going to try to give equivalent formulations of “injective”, “surjective”, and “bijective” that don't appeal to the normal set definitions.

**Definition.** Let  $G$  and  $H$  be groups and  $\phi : H \rightarrow G$  be a homomorphism. We say that  $\phi$  is an...

- (1) *Monomorphism* if, whenever  $\psi_1, \psi_2 : K \rightarrow H$  are homomorphisms with  $\phi \circ \psi_1 = \phi \circ \psi_2$ , we have that  $\psi_1 = \psi_2$ .
- (2) *Epimorphism* if, whenever  $\psi_1, \psi_2 : G \rightarrow K$  are homomorphisms with  $\psi_1 \circ \phi = \psi_2 \circ \phi$ , we have that  $\psi_1 = \psi_2$ .
- (3) *Isomorphism* if there is a homomorphism  $\psi : G \rightarrow H$  with  $\phi \circ \psi = \text{id}_G$  and  $\psi \circ \phi = \text{id}_H$ .

These correspond to the notions of “injective”, “surjective”, and “bijective” respectively.

*Exercise.* Suppose  $G$  and  $H$  are just sets, and drop the condition in the above definition that all maps are homomorphisms. Prove that monomorphism is equivalent to injective and that epimorphism is equivalent to surjective.

*Exercise.* Suppose that  $G$  and  $H$  are topological spaces, and replace every occurrence of the word “homomorphism” with “continuous map”. Prove that monomorphism is equivalent to injective and that epimorphism is equivalent to surjective. Prove that isomorphism is not equivalent to bijective (i.e. that being continuous with continuous inverse is strictly stronger than being bijective and continuous).

We'll now show these equivalences for our setting of groups.

**Proposition.** *Let  $\phi : H \rightarrow G$  be a homomorphism. Then  $\phi$  is a monomorphism if and only if it is injective.*

*Proof.* Suppose first that  $\phi$  is injective, and let  $\psi_1, \psi_2 : K \rightarrow G$  with  $\phi \circ \psi_1 = \phi \circ \psi_2$ . For any  $k \in K$ ,  $\phi(\psi_1(k)) = \phi(\psi_2(k))$ . But  $\phi$  is injective, so  $\psi_1(k) = \psi_2(k)$  and  $\phi$  is a monomorphism.

Conversely, suppose that  $\phi$  is a monomorphism, and suppose that  $\phi(h_1) = \phi(h_2) = g$ . Set  $K = \mathbb{Z}$  and define  $\psi_i : n \mapsto h_i^n$ . Then,  $\phi \circ \psi_i(n) = \phi(h_i^n) = \phi(h_i)^n = g^n$ . Thus,  $\phi \circ \psi_1 = \phi \circ \psi_2$ , so that  $\psi_1 = \psi_2$  by the definition of a monomorphism. Thus,  $h_1 = \psi_1(1) = \psi_2(1) = h_2$ , and so  $\phi$  is injective. ■

This notion of “monomorphism if and only if injective” turns out to be true in almost every situation (and every situation we will examine in this class). I may give a category theoretic proof of why later (this is related to the existence of free groups: whenever a “free” object exists, we can do this kind of construction - note that  $\mathbb{Z}$  is the free group on one object).

By contrast, this next notion of “epimorphism if and only if surjective” is not generally going to be true. Hopefully, the level of convolution of this proof will convince you of this fact.

**Proposition.** *Let  $\phi : H \rightarrow G$  be a homomorphism. Then  $\phi$  is an epimorphism if and only if it is surjective.*

*Proof.* Note that  $\phi$  being an epimorphism means that maps out of  $G$  are determined by how they act on  $\text{im } \phi$  (i.e. maps agreeing on  $\text{im } \phi$  must agree). In particular, when  $\text{im } \phi = G$ , this is clear.

Conversely, suppose that  $\phi$  is an epimorphism. Write  $H = \text{im } \phi \leq G$  (this is abusing notation, but we will never think of  $H$  as the domain of  $\phi$ ). We want to show that  $H = G$ , and we know that maps out of  $G$  are determined by how they act on  $H$ . Since an action  $G \curvearrowright X$  is the same as a map  $G \rightarrow S_X$ , group actions are also determined by how they act on  $H$ .

Now, the action  $G \curvearrowright G/H$  via  $g \cdot g'H = (gg')H$  is trivial if and only if  $H = G$  (since otherwise  $H$  has a nontrivial coset). Moreover, it suffices for  $H$  to be a fixed point of this action. Our goal will therefore be to find two actions on some set  $X \supseteq G/H$ : one where  $G$  acts on the subset  $G/H$  by left multiplication, and another where  $G$  fixes  $H \in G/H$  and where  $H$  still acts via left multiplication.

To do this, we introduce a dummy element  $*$ , and let  $X = G/H \sqcup \{*\}$ . Our first action will act  $G$  on  $G/H$  via left multiplication and fix  $*$ . We'll call  $\phi_1 : G \rightarrow S_X$  the induced homomorphism of this action. Our second needs to still be the same as left multiplication for elements of  $H$ . Since elements of  $H$  act trivially on both  $H$  and  $*$ , we can define our second action via

$$\phi_2(g) = \sigma \circ \phi_1 \circ \sigma,$$

where  $\sigma$  swaps  $H$  and  $*$  (and leaves all other elements fixed). Since any  $h \in H$  fixes both  $H$  and  $*$ , we have that  $\phi_2(h) = \phi_1(h)$ . Thus,  $\phi_2 = \phi_1$ , and so for any  $g \in G$ ,

$$H = \phi_2(g)(H) = \phi_1(g)(H) = gH.$$

Thus,  $H = G$ . ■

*Exercise.* Show that a homomorphism is an isomorphism if and only if it is bijective. That is, if  $\phi$  is a bijective homomorphism, show that  $\phi^{-1}$  is also a homomorphism.